Bioinformatics File Formats

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Learning Objectives

- Understand what bioinformatics file formats are in bioinformatics.
- Learn about three common file formats: FASTA, VCF, BAM.
- Be able to identify which format is used for what kind of biological data.
- See how standardization helps researchers share and reproduce results.

What is a File Format?

- A **file format** is a way data is stored in a file, it tells software how to read and write biological information.

- Computers don't "see" DNA as ACGT, **they see files**.

- Scientists around the world need **a shared way** to store and share data.

Overview of Common File Formats

Format	Stores	File Extension
FASTA	DNA/Protein Sequences	.fasta, .fa
VCF	Genetic Variants	.vcf
SAM	Sequence Alignment	.sam
BAM	Sequence Alignment	.bam (binary)

What is a FASTA File?

- **FASTA** is a text-based format for storing **nucleotide** or **protein** sequences.
- Usually nd with the extension **.fasta**

- Each FASTA entry has:

- Line 1: A header line starting with ">“
- Line 2+: One or more sequence lines

Structure of A FASTA File

- >sequence_1 description
ATCGATCGATCGATCG

- >sequence_2 description
GCTAGCTAGCTA

FASTA File: Example

- >AL645882.2 Streptomyces coelicolor A3(2) complete genome
  CCCGCGGAGCGGGTACCACATCGCTGCGCGATGTGCGAGC
  GAACACCCGGGCTGCGCCCGGGTGTTGCGCTCCCGCTCCG
  CGGGAGCGCTGGCGGGACGCTGCGCGTCCCGCTCACCAAG
  CCCGCTTCGCGGGCTTGGTG

FASTA File: Example (2)

- >EE884703.1 sp_e13_42 Sweetpotato cDNA library E Ipomoea batatas cDNA 5', mRNA sequence
  GGGGTATATGGCTTGGATATATAGCTATTAAAGATTTTACCTAT
  GGCATATTTCAATGTGTATTAGTACTAAGTAAGAATGATTGCA
  AGGTGTATTAACTACAAATATTGCAATAAAAGTCCCTGTTACC
  AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA

Importance of FASTA files?

- Most basic and widely-used format.
- **Used in:**
  - Genome assemblies.
  - BLAST searches.
  - Protein databases.
- Works across platforms and tools.

What is a SAM File?

- **SAM** = Sequence Alignment/Map.
- A text-based format for storing **aligned sequence** data.
- Stores **aligned sequences** mapped to a reference genome.
- Tool: **SAMtools**

Structure of a SAM File

- **Contains:**
  - **Header section:** starts with **@** (e.g., @SQ, @PG).
  - **Alignment section**: each line = 1 read.

SAM File - Example

```
@HD    VN:1.6    SO:coordinate
@SQ    SN:chr1   LN:248956422
r001   0         chr1   7   60   8M   *   0   0   TTAGATA   *
r002   0         chr1   9   60   8M   *   0   0   CAGATAA   *
```

Key Fields

- **QNAME**: query name (read ID)
- **FLAG**: bitwise flag (alignment info)
- **RNAME**: reference sequence name
- **POS**: position of alignment
- **MAPQ**: mapping quality
- **CIGAR**: alignment operations

Key Fields

- **RNEXT**: Reference name of the next read.
- **PNEXT**: primary alignment position.
- **TLEN**: observed template length.
- **SEQ**: segment sequence.
- **QUAL**: Base quality.

Importance of SAM files?

- Tracks where tumor vs. normal reads align.
- Used in variant calling (mutations, SNPs, indels).
- Works with tools like BWA, Bowtie, SAMtools.
- Essential for reproducibility and transparency.

What is a BAM File?

- **BAM** = Binary Alignment Map.
- Compressed, binary version of SAM (Sequence Alignment Map).
- Stores **aligned sequences** mapped to a reference genome.

Importance of BAM files?

- Visualisation of alignments
- Variant calling
- RNA-seq quantification
- Compressed = storage-efficient

What is a VCF File?

- **VCF** = Variant Call Format
- Stores **variations (mutations)** in DNA between individuals.
-

CHROM	POS	ID	REF	ALT
1	1234	rs1	A	G

Structure of a VCF File

- **Components:**
  1. **Header lines** (starts with ##) → Metadata.
  2. **Column header** (starts with #CHROM) → Labels for each field. **8 key fields.**
  3. **Data rows** → The variants

VCF File – Structure

- **##fileformat**=VCFv4.2

- **#CHROM POS ID REF ALT QUAL FILTER INFO**

  1 123456 . A G 50 PASS .

VCF File – Key Fields

- **#CHROM:** Chromosome identifier.
- **POS**: Reference position.
- **ID**: Unique identifiers
- **REF**: Reference base.
- **ALT**: Alternate base(s).

VCF File – Key Fields

- **QUAL**: Quality score
- **FILTER**: Indicates which filters have failed, PASS or MISSING
- **INFO** – Site-level information
- **FORMAT** – Sample-level field name declarations
- **<SAMPLE DATA>** – Sample-level field data

Importance of VCF files?

- **Gold standard** for storing SNPs, indels and sharing variant data.
- Used by databases like gnomAD, 1000 Genomes, ClinVar.
- Compatible with tools like bcftools, GATK and PLINK.

Summary Table

Format	Data Type	Used For	Tools
FASTA	Sequences	BLAST, genome reference	BLAST, BWA
VCF	Variants	Genotyping, GWAS	GATK, bcftools
BAM	Alignments	RNA-seq, mapping	samtools, IGV

Importance of Standard Formats

- **Reproducibility**: Other scientists can repeat your analysis
- **Data sharing**: Easier collaboration
- **Tool compatibility**: Saves time

Practice Questions

1. What type of data is stored in a FASTA file?
2. Why do we use BAM instead of SAM?
3. What do REF and ALT mean in a VCF file?

Conclusion

- The program ensures that students gain hands-on experience, **technical expertise** and **scientific reasoning** skills needed to excel in **bioinformatics** and **computational biology**.

Coming up next…

Major Bioinformatics Databases:

- ✓ TCGA
- ✓ Ensembl
- ✓ UCSC Genome Browser

Further Reading/ References

- EMBL-EBI (2025). Human genetic variation. Accessed from https://www.ebi.ac.uk/training/online/courses/human-genetic-variation-introduction/variant-identification-and-analysis/understanding-vcf-format/